Design and Implementation of the Titan Eye+ Eyewear Retail

CHAPTER 1

1.1 Introduction

The Titan Eye+ Eyewear Retail Database is designed to represent the business operations of one of India's leading eyewear retail brands. Titan Eye+ offers a wide range of optical products including prescription eyeglasses, sunglasses, contact lenses, blue-light protection glasses, computer glasses, reading glasses, and eyewear accessories. The company also provides professional eye testing services through its retail stores and online platform.

As Titan Eye+ serves millions of customers across India through its physical stores and e-commerce platform, it generates a huge volume of business data every day. This data includes customer details, product information, brands, categories, inventory, suppliers, eye test records, orders, payments, delivery status, reviews, and warranties. Managing such a large amount of information manually is difficult and may lead to duplicate records, inconsistent data, delays in processing, and poor customer service.

To overcome these challenges, a well-designed relational database is essential. The database stores all business information in a structured manner and establishes relationships between different entities using Primary Keys and Foreign Keys. It also applies normalization techniques to eliminate redundancy and maintain data integrity.

This project focuses on designing and implementing the Titan Eye+ Eyewear Retail Database using Oracle SQL. The database supports essential business activities such as customer registration, product management, eye test appointment booking, inventory management, order processing, payment handling, warranty tracking, and customer feedback. It demonstrates how database management systems help improve operational efficiency, ensure accurate data storage, and support decision-making in a modern eyewear retail business.

1.2 Problem Statement

Titan Eye+ manages a large amount of business information every day. Customer details, product catalogs, eye test records, inventory, suppliers, payments, warranties, and order history must be stored accurately. If this information is managed manually or using unorganized files, several problems can occur such as data redundancy, inconsistent records, loss of information, slow retrieval, inventory errors, and poor customer service.

The company requires a centralized relational database that can securely manage all business operations while maintaining data consistency. The database should establish relationships between various entities, reduce duplicate data through normalization, and provide fast retrieval of information using SQL queries.

The proposed Oracle SQL database addresses these challenges by providing an efficient, reliable, and scalable solution for managing the complete business operations of Titan Eye+.

1.3 Objectives

The main objectives of the project are:

To design a relational database for Titan Eye+.

To identify all major business entities such as Customers, Products, Brands, Categories, Eye Tests, Orders, Payments, Inventory, Suppliers, Reviews, and Warranty.

To establish relationships among entities using Primary Keys and Foreign Keys.

To apply normalization techniques to reduce redundancy.

To maintain data consistency and integrity.

To implement constraints such as PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, and CHECK.

To support CRUD operations.

To manage customer orders efficiently.

To maintain accurate inventory records.

To store eye examination details.

To track warranty information.

To generate reports for business analysis.

To gain practical knowledge in Oracle SQL database design.

1.4 Motive

The primary motive of this project is to understand how a real-world retail organization manages its business data through a relational database. Titan Eye+ operates numerous retail stores and serves thousands of customers every day.

Efficient management of customer information, product inventory, eye test appointments, orders, and payments is essential for smooth business operations.

This project helps in applying theoretical database concepts into practical implementation using Oracle SQL. It provides experience in designing database tables, defining relationships, implementing constraints, and performing SQL operations. The project also demonstrates how database systems improve business efficiency, minimize data redundancy, enhance customer service, and support future business growth.

1.5 Stakeholders

The stakeholders involved in the Titan Eye+ database include:

Customers

Database Administrator (DBA)

Store Manager

Sales Executive

Optometrist

Inventory Manager

Supplier

Customer Support Team

Finance Department

Management Team

Each stakeholder interacts with the database according to their responsibilities, ensuring efficient management of daily business activities.

1.6 Business Requirements

The database should support the following business operation

Store customer information securely.

Maintain product details including frames, lenses, sunglasses, and contact lenses.

Manage brands and product categories.

Schedule eye testing appointments.

Store eye examination results.

Process customer orders.

Record payment transactions.

Manage product inventory.

Track supplier information.

Maintain warranty records.

Store customer reviews and ratings.

Maintain customer wishlists.

Generate sales and inventory reports.

Ensure accurate stock management.

Support multiple retail store operations.

Provide efficient product searching.

Maintain historical customer purchase records.

1.7 Functional Requirements

The system should be capable of performing the following functions:

Customer registration.

Customer login.

Profile management.

Product management.

Brand management.

Category management.

Inventory management.

Supplier management.

Eye test appointment booking.

Eye prescription storage.

Shopping cart management.

Order placement.

Payment processing.

Invoice generation.

Warranty tracking.

Product review submission.

Wishlist management.

Product search.

Order tracking.

Report generation.

CRUD operations for all entities.

Data retrieval using SQL queries.

Relationship management using foreign keys.

1.8 Non-Functional Requirements

Security :

- √ Customer and payment information should be protected.
- √ Only authorized users can modify sensitive records.
- √ Authentication and authorization should be implemented.
- √ Database constraints should ensure data integrity.

Reliability :

- √ Data should remain accurate and consistent.
- √ Backup and recovery mechanisms should be available.
- √ Transactions should maintain consistency.

Performance :

- √ SQL queries should execute quickly.
- √ The database should efficiently handle thousands of records.
- √ Reports should be generated with minimum delay.

Scalability :

The database should support future expansion.

- √ New products, stores, and customers should be added without major structural changes.
- √ Availability

- √ The system should remain available throughout business hours.
- √ Downtime should be minimized.
- √ Maintainability
- √ Database tables should be easy to modify.
- √ Proper normalization should simplify future maintenance.
- √ Documentation should support future development.
- √ Usability
- √ The system should be easy to understand and operate.
- √ Reports should be clear and meaningful.
- √ Portability
- √ The database should be deployable on different Oracle environments.

1.9 Conclusion :

The Titan Eye+ Eyewear Retail Database project demonstrates the design and implementation of a well-structured relational database using Oracle SQL. The database efficiently manages customer information, products, brands, inventory, eye test records, suppliers, orders, payments, warranties, and customer feedback. By applying normalization, database constraints, and relational concepts, the system ensures data integrity, minimizes redundancy, and improves business efficiency.

The project reflects how a real-world eyewear retail organization manages its operations using a centralized database management system. It provides practical experience in database design, SQL implementation, and business data management while preparing students for real-world database development projects.